PRESENTING

Cluco

AI Agent Observability Platform

Pradeep Reddy Gorla

Senior Product Manager (AI/GenAI) · Platform & Applied ML

Tracing · Evaluation · Cost Analytics · Live Monitor · Agent Flow

March 2026

INTRODUCTION

My Journey

2016
IIT (BHU)

B.Tech, Mining Engineering · GPA 8.7
Developed an application to predict mine accidents using sensor data and statistical models. The turning point that sparked my passion for launching software solutions.

Data Modeling · Systems Thinking

2016–18
Virtusa / BT

Software Engineer
Products: Observability Platform (Prometheus, Grafana, ELK) for 1M+ devices · API Modernization (REST, GraphQL, Kafka) · Real-time Fault Notification System

Java / Spring Boot · Kafka · Docker / CI-CD

2018–22
Virtusa / BT

Product Manager, Technology
Products: Predictive Maintenance (LSTM + Isolation Forest) · Churn Prediction · Convergent Billing · Cloud Migration (monolith to microservices) · OSS/BSS (150+ releases)
PM Skills: PRDs · P&L (£45M+) · C-level stakeholder mgmt · ROI cases · Agile with 35+ engineers

2023
Raymond James

PM Intern
Product: Investment Research Chatbot MVP / multi-agent RAG on AWS (LangChain, LangGraph, OpenAI). 75% analyst adoption in 3 weeks.
PM Skills: User research with equity analysts · MVP scoping · Product-market fit

2024
UNC Kenan-Flagler

MBA · Beta Gamma Sigma Honor Society
Capital Markets · Business Analytics · Technology & Innovation Management · KFBS Merit Fellow

2024–25
HubSpot

Product Manager, AI/ML
Products: Ads-to-CRM Attribution · Multi-Agent Ad Creative Generator · Multimodal AI Moderation (CNN + ViT + LLaMA-Vision)
PM Skills: Technical PRDs · ML requirements · A/B testing frameworks

2025+
Wayfair

Senior Product Manager
Products: Programmatic Ads Platform ($30M) · Conversational AI Agent (MCP) · Shared Embedding Platform · LLM Recommendation Agent
PM Skills: 50+ A/B tests · OKRs · Platform strategy · Cross-org alignment

PORTFOLIO

My AI Projects

Cluco

AI Agent Observability Platform with traces, LLM-as-Judge evaluation, cost analytics, and closed-loop quality improvement.

Python / FastAPI / React / MongoDB

This Demo

TelcoAssist

Multi-agent telecom customer support with LangGraph routing, Pinecone RAG, and full Cluco instrumentation.

Python / LangGraph / Pinecone

5 Agents

TravelMind

Production-grade 8-agent AI travel assistant orchestrated via LangGraph with 8 external APIs + RAG.

Python / LangGraph / 8 APIs

8 Agents

Text-to-SQL

Self-optimizing NLP engine using 4 patterns: Intent Detection, RAG, Self-Correction, Optimization.

Python / OpenAI / PostgreSQL

NLP

Demand Draft AI

20-agent legal document pipeline reducing 40-hour manual process to 1-hour AI-assisted workflow.

Python / LangGraph / Neo4j

20 Agents

5
AI Projects Built

50+
Total Agents Deployed

3
Instrumented with Cluco

MARKET RESEARCH

The LLM Observability Market

$673M	$8-17B	31-34%	38%
2025 Market Size	Projected 2033-34	CAGR Growth	North America Share

Competitive Landscape – Where the Gaps Are

Capability	Langfuse	LangSmith	Braintrust	Arize	Datadog	Cluco
Distributed Tracing	✓	✓	✓	✓	✓	✓
LLM-as-Judge Eval	✓	✓	✓	●	○	✓
Cost Analytics	✓	●	○	●	✓	✓
Live WebSocket Monitor	○	○	○	○	●	✓
Closed-Loop Quality	●	●	●	○	○	✓
Prompt Registry	✓	●	✓	○	○	✓
Agent Flow Graph	●	✓	○	○	○	✓
Self-Hosted + OSS	✓	○	○	✓	○	✓

✓ Full ● Partial ○ None

Market White Space – No Competitor Fully Addresses

Closed-loop quality cycle: No tool automates trace to eval to anomaly to alert to fix in one platform

Real-time WebSocket debugging: Most tools are batch/polling only, no live span streaming

Product-agnostic multi-tenancy: Most tools are framework-locked (LangSmith) or enterprise-only (Datadog)

Self-hosted + full-stack: Open-source tools lack evaluation depth; enterprise tools don't self-host

STRATEGIC ANALYSIS

SWOT Analysis

Strengths

- Full-stack platform: tracing + evaluation + cost + live monitor
- Closed-loop quality cycle (unique differentiator)
- Product-agnostic / multi-tenant architecture
- Python SDK with decorator-based instrumentation
- Real-time WebSocket streaming for live debugging
- Self-hosted / data stays in your infrastructure

Weaknesses

- Single-developer project / limited team bandwidth
- No enterprise support or SLA guarantees yet
- Limited framework integrations vs competitors
- No built-in authentication / RBAC system
- MongoDB-only storage (no Postgres/ClickHouse)
- Early-stage documentation and community

Opportunities

- $17B TAM by 2034 with 31-34% CAGR growth
- EU AI Act driving compliance-focused observability
- Multi-agent system adoption creating new complexity
- Enterprise demand for self-hosted / on-prem solutions
- OpenTelemetry gen_ai.* standards adoption wave
- Partnership potential with LangChain / LangGraph ecosystem

Threats

- Established players (Datadog $20B, LangSmith) adding LLM features
- Well-funded startups (Braintrust $80M, Langfuse $12M)
- OpenTelemetry native tooling reducing need for specialized platforms
- Cloud provider lock-in (AWS Bedrock, Azure AI adding tracing)
- Rapid commoditization of basic tracing features
- Open-source competitors with larger contributor base

STRATEGIC ANALYSIS

Porter's Five Forces

Threat of New Entrants

HIGH

• Low technical barriers with OSS tooling (OTel, LiteLLM)
• Any startup can build basic tracing in weeks
• OpenTelemetry gen_ai.* standards accelerating entry

Cluco: Differentiate via closed-loop quality, not just tracing

Supplier Power

LOW

• LLM providers (OpenAI, Anthropic, Google) interchangeable
• No single vendor controls the observability layer
• MongoDB / storage is commodity infrastructure

Cluco: Remain model-agnostic, support all providers

Industry Rivalry HIGH

Buyer Power

MEDIUM-HIGH

• 15+ alternatives available, low switching costs
• Open source reduces vendor lock-in significantly
• Enterprise buyers demand proof before committing

Cluco: Self-hosted + data portability + OSS transparency

Threat of Substitutes

MEDIUM

• DIY logging (print statements, custom dashboards) still common
• Raw OpenTelemetry collectors without specialized UI
• Cloud-native tools: AWS Bedrock logs, Azure AI tracing

Cluco: Value beyond raw data - evaluation, alerting, optimization

Competitive Rivalry

HIGH

15+ funded competitors converging on features. Langfuse ($12M), Braintrust ($80M), Datadog ($20B+). Feature parity on tracing is accelerating rapidly.

Cluco: Win on full-stack integration depth

MARKET RESEARCH

Personas & Pain Points

Who uses LLM Observability?

ML / LLM Engineer
Instruments traces, debugs agent behavior, configures prompts, runs evaluations

"Why did the agent make this decision?"

AI Product Manager
Defines success metrics, designs eval criteria, navigates non-deterministic outcomes

"Is the agent meeting our success criteria?"

Platform / LLMOps Engineer
Manages prompt versions, monitoring pipelines, cost control, governance, guardrails

"How do I manage cost, latency, and versions?"

QA / Domain Expert
Creates annotations, writes test scenarios, decides release readiness for AI outputs

"Is this output correct? Can we ship it?"

Top Pain Points They Face

Silent Failures
HTTP 200 OK with hallucinated output. Traditional APM shows everything is fine while the model confidently returns wrong answers.

Demo-to-Production Gap
Agent works in demo, breaks in production. Integration complexity can cost $500K+ in wasted salary across teams.

Evaluation Is Unsolved
No consensus on scoring. 89% have observability, only 52% run proper evaluations, just 12% automate them.

Cost Explosion
Agents making 47 API calls for simple queries. Minor prompt edits cause +$300-600/month. No per-request cost tracking.

Model Drift
Provider model updates silently change behavior without any code change. Prompt drift degrades quality over weeks.

MARKET RESEARCH

Adoption Journey

Week 0
Incident

Production hallucination triggers awareness. Traditional APM shows no errors. Team realizes they are debugging blind.

"We have no idea why the agent said that."

N/A

Week 1-2
Instrument

Add SDK (5-15 min setup). Start capturing inputs, outputs, tokens, cost, latency for every LLM call.

"Now we can see what the model received."

SDK · Tracing

Week 2-4
Debug

Use traces to debug failures. Inspect full execution chains. Debugging time drops 73%.

"Found the retriever returning wrong chunks."

Traces · Spans

Month 1-2
Evaluate

Label 20-50 production traces. Build baseline eval datasets. Set up LLM-as-judge scoring.

"We finally have a quality baseline."

Eval Engine · Datasets

Month 2-3
Experiment

Prompt registry + versioning. A/B experiments comparing prompt variants against eval datasets.

"Prompt v3 scores 18% higher."

Prompts · Experiments

Month 3-6
Dashboards

Business dashboards for stakeholders. Cost attribution by user/feature/model. Alert rules.

"Leadership sees AI ROI metrics."

Cost · Alerts

Month 6+
Mature Ops

CI/CD eval gates. Continuous online evaluation. Compliance audit trails automated.

"No prompt ships without eval gates."

CI/CD · Compliance

THE PROBLEM

Why LLMs Need Different Monitoring

Traditional Web Monitoring

Status: 200 OK ✓

✓ Error Rate: Zero

✓ Latency: Normal (1.2s)

✓ Throughput: Healthy (120 rps)

✓ Memory: Stable

But the model returned:

"Australia's capital is Tokyo"

with 0.97 confidence score

Superficial Success = Hidden Quality Failures

VS

LLM Observability

1. **Exact Prompt Sent**
   System messages, injected context, temperature
2. **Retriever & Similarity**
   Retrieved docs, similarity scores, chunk metadata
3. **Model Output & Confidence**
   Full completion text with confidence metrics
4. **Token Count & Cost**
   Per-request: 2,340 tokens / $0.047
5. **Per-Step Latency Breakdown**
   Prompt 3ms / Retrieval 180ms / Model 650ms / Tool 120ms
6. **Quality Evaluation Scores**
   Relevance: 0.92 / Faithfulness: 0.34 / Groundedness: 0.41

AGENT LIFECYCLE

Agent Development Lifecycle

- **New use case**
- Build a working prototype that passes basic quality "vibe checks"
- Test drive with human testers
- Identify quality issues and create repros
- Fix issues in agent code
- Test fixes against evaluation datasets
- **Get stakeholder sign-off for production**
- Release Agent to production & monitor quality

↺ Loop back to "Identify quality issues"

Observability Features by Stage

- Tracing & Instrumentation
- Labeling & Annotation
- LLM Judges (Automated)
- Evaluation UIs
- Evaluation Datasets
- Prompt Registry
- Prompt Optimization
- AI/BI Dashboards
- Experiment Tracking

AGENT LIFECYCLE

Cluco Across the Lifecycle

Build Prototype

SDK instrumentation from day one. Prompt Registry for versioning. Trace every LLM call, tool invocation, and retrieval step automatically.

Test with Human Testers

Annotation Queues for manual review. Labeling Sessions with configurable score configs. Public review routes for external testers.

Identify Quality Issues

LLM-as-Judge automated scoring. Anomaly Detection with statistical baselines. Full trace inspection with prompt/completion capture.

Fix Issues in Agent Code

Prompt Optimizer suggests improvements. AI Debug Assistant on traces. Trace Comparison for side-by-side regression checking.

Test Fixes

Evaluation Datasets curated from production traces. Experiments to compare prompt/model variants. Evaluation Suites for grouped metrics.

Stakeholder Sign-off

Quality Trends dashboards with historical data. Experiment results as evidence for decisions. AI/BI Dashboards for business metrics.

Release & Monitor Quality in Production

Live Monitor (WebSocket streaming) · Agent Flow visualization · Cost Analytics per-request · Email alerts + rules engine · Auto Judge Monitors on every trace · Scheduled Evaluations · Anomaly Detection with baselines

CLUCO PLATFORM

Architecture Overview

Product-agnostic observability for AI agents. Multi-tenant by `product_id` + `service_name`.

CLIENT LAYER

AI Applications — LangChain, LangGraph, Custom Agents → **cluco_obs SDK** — Python decorators + callbacks + LangChain handler → **HTTP / WebSocket / SSE** — Batch ingest + streaming + events

↓

SERVER LAYER (FASTAPI / PORT 9410)

- **Trace API** — Ingest + query
- **Eval Engine** — LLM Judge + scoring
- **Alert Engine** — Rules + email
- **Prompt Registry** — Versions + optimizer
- **WS Server** — Live streaming

↓

STORAGE & PRESENTATION

MongoDB — Traces, Spans, Evals, Prompts, Alerts → **React UI (Port 9411)** — 36 pages: Dashboard, Traces, Evals, Cost, Live Monitor, Agent Flow, Prompts, Experiments

CLUCO PLATFORM

Integration Architecture (UML)

YOUR AI APPLICATION

LangChain / LangGraph App
Chains, agents, retrievers, tools

Custom Python Agent
Direct OpenAI / Anthropic calls

RAG Pipeline
Embeddings + vector search + LLM

```
@tracer.trace()
ClucoHandler()
```

CLUCO PLATFORM

cluco_obs SDK
Decorators, callbacks, auto-capture spans

↓ POST /api/v1/traces/ingest

REST API (FastAPI)
Trace ingest, span stream, eval scoring

↓ Store + Process

MongoDB	WS Server
Eval Engine	Alerts

OUTPUTS

React Dashboard
36 pages: traces, agent flow, cost, evals, live monitor, prompts

Email Alerts
Anomaly detection, quality drops, cost thresholds

REST / WS API
Programmatic access for CI/CD gates, custom dashboards

Quality Reports
Eval trends, experiment results, stakeholder evidence

CLUCO PLATFORM

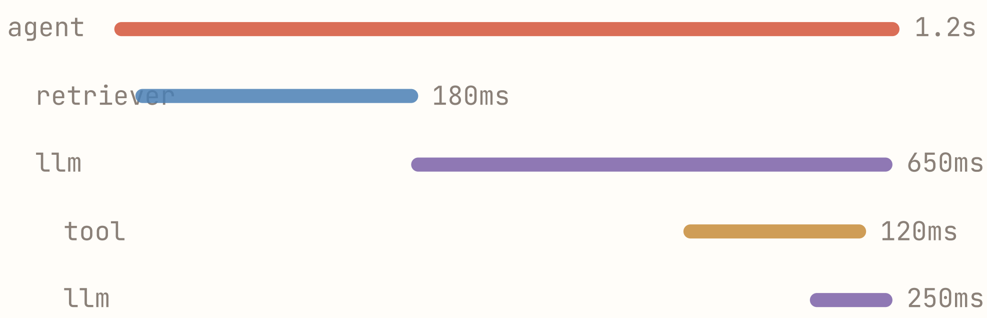
How Cluco Solves Each Failure Mode

Prompt Drift →	**Prompt Registry + Versioning + Optimizer** Track every prompt version, compare outputs across versions, auto-optimize
Silent Retrieval →	**Full Trace Inspection (Retriever Spans)** See exact chunks retrieved with similarity scores, inspect every span's I/O
Cost Explosion →	**Cost Analytics Dashboard + Budget Alerts** Per-model, per-request cost tracking with trends, alerts, and token breakdowns
Latency Anomalies →	**Gantt Timeline + Anomaly Detection** Per-step latency visualization, statistical baselines, category alerting
Agent Loops →	**Agent Flow Graph + Live Monitor** Real-time WebSocket streaming, visual flow graph, detect infinite cycles

CLUCO PLATFORM

Traces & Spans

Gantt-Style Timeline



Span Kinds:

LLM | Tool | Retriever | Embedding | Agent | Chain

What Each Span Captures

LLM Span

Model, prompt_messages, completion, tokens (in/out), cost, latency, temperature

Retriever Span

Query text, retrieved documents, similarity scores, chunk metadata, vector DB source

Tool Span

Tool name, JSON arguments, result payload, execution time, error state, retries

Embedding Span

Model, input text, vector dimensions, token count, batch size

Trace Comparison

Compare traces side-by-side to debug regressions. Diff on spans, inputs, outputs, and timing.

CLUCO PLATFORM

LLM Calls & Cost Analytics

LLM Calls View

Model	Tokens	Cost	Latency
gpt-4o	2,340	$0.047	1.2s
gpt-4o-mini	890	$0.002	0.4s
gpt-4o	3,120	$0.062	1.8s
claude-3.5	1,560	$0.023	0.9s

Full Prompt & Completion Capture

Every LLM call stores complete prompt_messages and completion in MongoDB. Nothing is sampled or dropped.

Cost Analytics Dashboard

$127
This Month

359K
Total Tokens

1.4K
LLM Calls

- Cost trends over time (daily / weekly / monthly)
- Per-model cost breakdown with percentages
- Per-request cost tracking and anomaly flags
- Budget alerts with configurable thresholds
- Token usage breakdown by span type

CLUCO PLATFORM

Evaluation Engine

Three Sources of Quality Scores

LLM-as-Judge (Automated)

Scores traces on relevance, faithfulness, groundedness, helpfulness. Runs on trace completion via auto judge monitors. Configurable criteria.

Human Reviewer

Annotation queues with labeling sessions. Configurable score configs (1-5, binary, categorical). Public review routes for external domain experts.

User Feedback

Direct user scores and assessments attached to traces. Thumbs up/down, star ratings, free-text comments feed into quality measurement.

Evaluation Capabilities

Datasets

Curate test sets from production traces

Experiments

Compare prompt / model variants A/B

Scheduled Evals

Automated recurring quality checks

Eval Suites

Group metrics into test suites

Quality Trends

Track quality over time with baselines

Auto Monitors

Judge runs on every trace completion

Closes the 89% → 52% gap

by making evaluation automatic, continuous, and integrated.

CLUCO PLATFORM

Live Monitor & Agent Flow

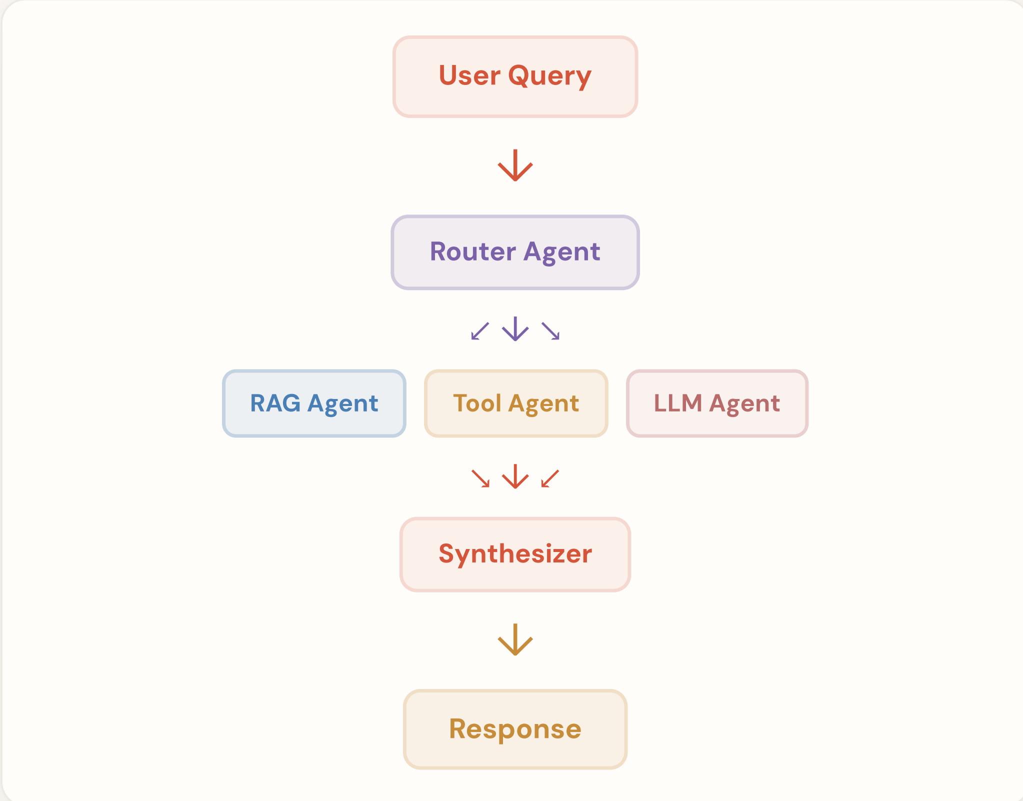
Real-Time Live Monitor

Live / WebSocket Streaming

- ▶ Watch traces execute in real-time as they happen
- ▶ See spans appear the moment they complete
- ▶ Stream LLM completions token-by-token
- ▶ Detect infinite agent loops instantly
- ▶ Active trace count and throughput metrics

```
// WebSocket endpoints
/ws/traces/{trace_id}/live
/ws/live
// SSE polling fallback
/api/v1/traces/{id}/events
```

Agent Flow Visualization



Built with **React Flow** + **dagre** auto-layout. Interactive: zoom, pan, click span for detail.

CLUCO PLATFORM

SDK & Integration

Python SDK: `cluco_obs`

```python
from cluco_obs import ClucoTracer

# Initialize tracer
tracer = ClucoTracer(
  api_url="https://cluco-backend.onrender.com",
  product_id="my-app"
)

# Decorator-based instrumentation
@tracer.trace(span_kind="llm")
def generate_response(prompt):
  return openai.chat(prompt)

# LangChain callback handler
from cluco_obs.langchain import ClucoHandler
chain.invoke(input, callbacks=[ClucoHandler()])
```

API Endpoints

`POST /api/v1/traces/ingest`
Batch ingest complete traces with all spans

`POST /api/v1/spans/stream`
Stream individual spans for live updates

`WS /ws/traces/{id}/live`
WebSocket for real-time trace streaming

`GET /api/v1/traces/{id}/events`
SSE for polling-based live updates

Supported Frameworks

LangChain · LangGraph · OpenAI · Custom · DSPy · CrewAI

DIFFERENTIATOR

Closed-Loop Quality Improvement

Cluco's key differentiator: a continuous, automated quality cycle that catches issues before users notice.

Trace Ingest
SDK captures everything
→
Auto Evaluation
LLM-as-Judge scores
→
Anomaly Detection
Statistical baselines
→
Alert
Email + rules engine
→
Debug
AI debug assistant
→
Optimize
Prompt optimizer
↺

Automatic
Judge monitors trigger on every trace

Continuous
Quality trends with statistical baselines

Actionable
Alert to fix, integrated in one platform

DEMO

Live Demo Walkthrough

1. **Dashboard Overview**
   Key metrics, trace counts, active agents, quality scores
2. **Traces List & Filters**
   Filter by product, service, status, date range, metadata
3. **Trace Detail View**
   Gantt timeline, hierarchical span tree, full I/O
4. **Agent Flow Graph**
   Interactive DAG of agent reasoning and tool calls
5. **Live Monitor**
   Watch traces execute in real-time via WebSocket
6. **Evaluations Hub**
   LLM-as-Judge scores, experiments, quality trends
7. **Cost Analytics**
   Per-model costs, daily trends, budget tracking
8. **Prompt Registry**
   Version history, diff comparison, optimization

UI: https://cluco-ui.onrender.com • Backend: https://cluco-backend.onrender.com

Thank You

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Questions?

Tracing · Evaluation · Cost Analytics · Live Monitor · Agent Flow